

The background of the image features a dark blue gradient on the left, transitioning into a large, vibrant, abstract shape on the right. This shape is composed of overlapping curved segments in shades of orange, pink, and purple, creating a dynamic, modern aesthetic.

AWS re:Invent

NOV. 27 – DEC. 1, 2023 | LAS VEGAS, NV

API 305

Building resilience in decoupled applications with dead-letter queues

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Principal Solutions Architect
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“

Everything fails all the time.

Dr. Werner Vogels

VP and CTO, Amazon.com

Overview of Amazon Simple Queue Service (Amazon SQS)

WHAT IT IS

Simple, flexible, fully managed **message queuing service** for **reliably** and continuously exchanging any volume of messages from anywhere

USE CASE

Build **decoupled**, highly scalable **microservices**, **distributed systems**, and **serverless applications** in the cloud

CAPABILITIES

- Nearly **infinite scalability** without pre-provisioning capacity
- **Low latency**
- Offers **standard** and **FIFO** queues

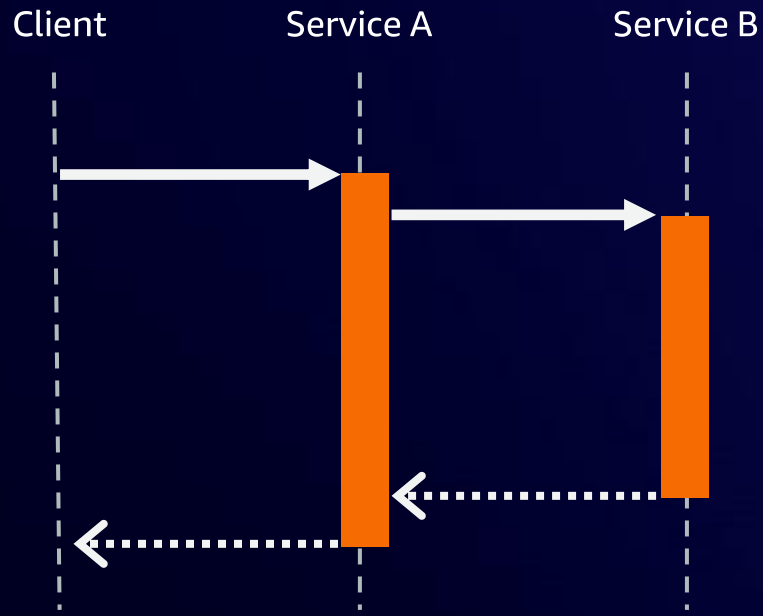


100M messages per sec

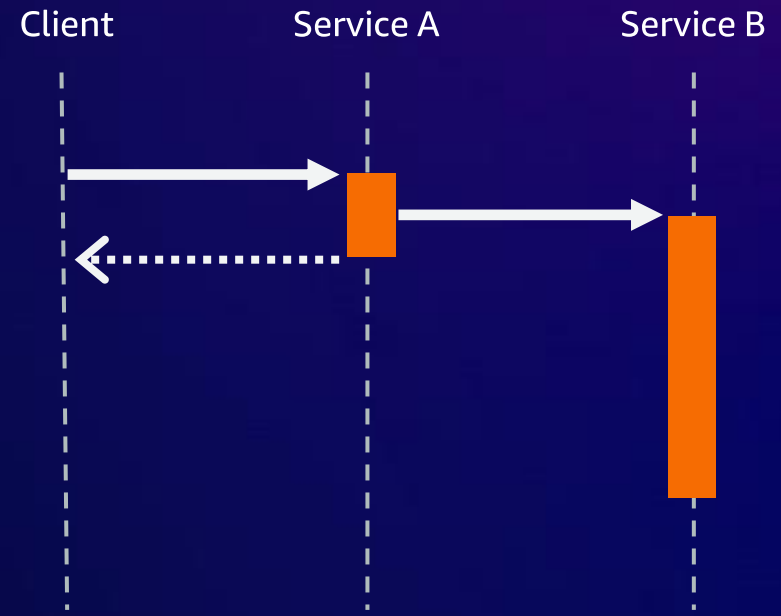
```
> aws sqs create-queue --queue-name MyQueue
```



Synchronous vs. asynchronous integration

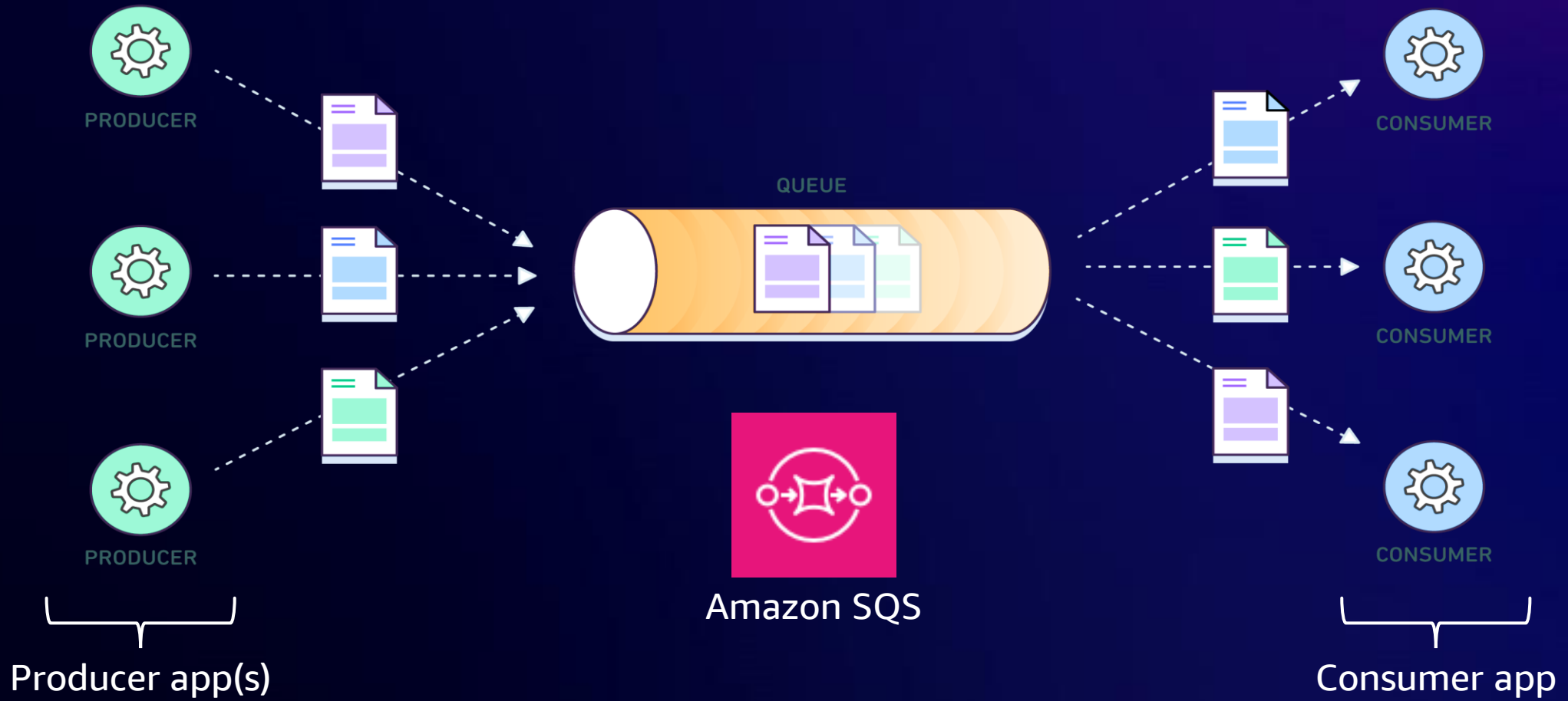


Synchronous
commands

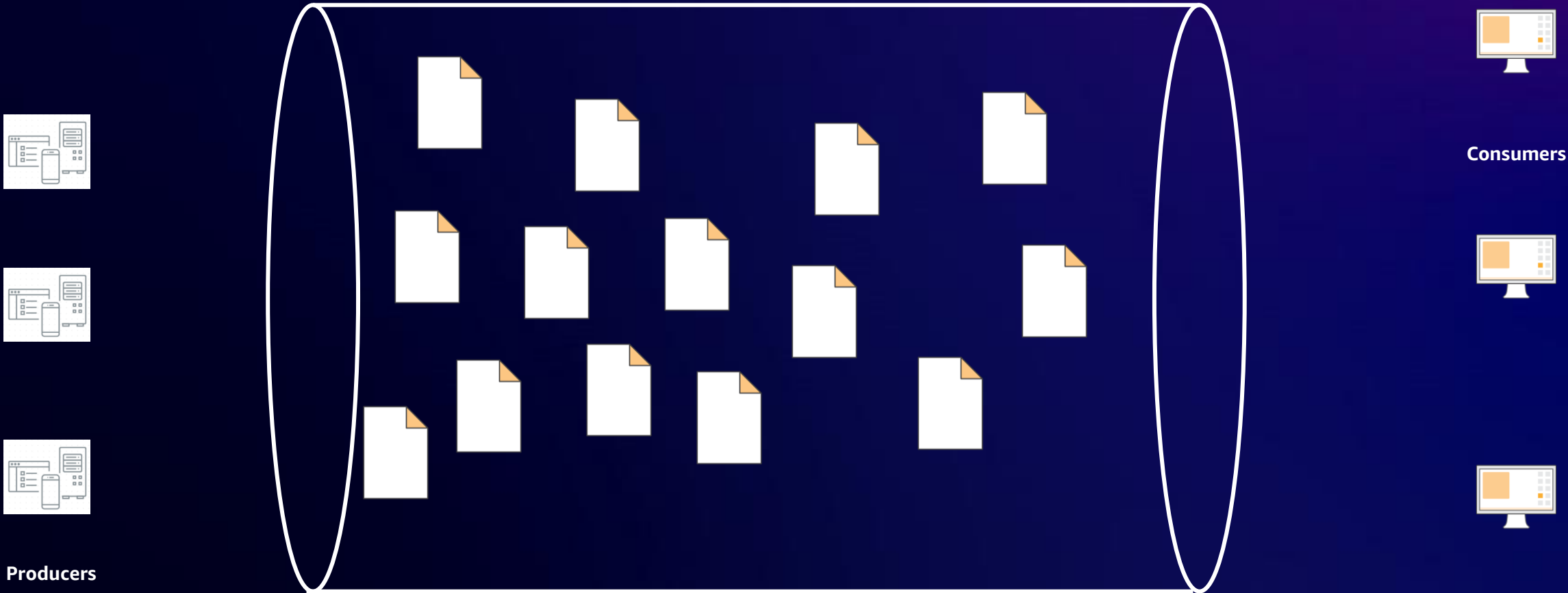


Asynchronous
events

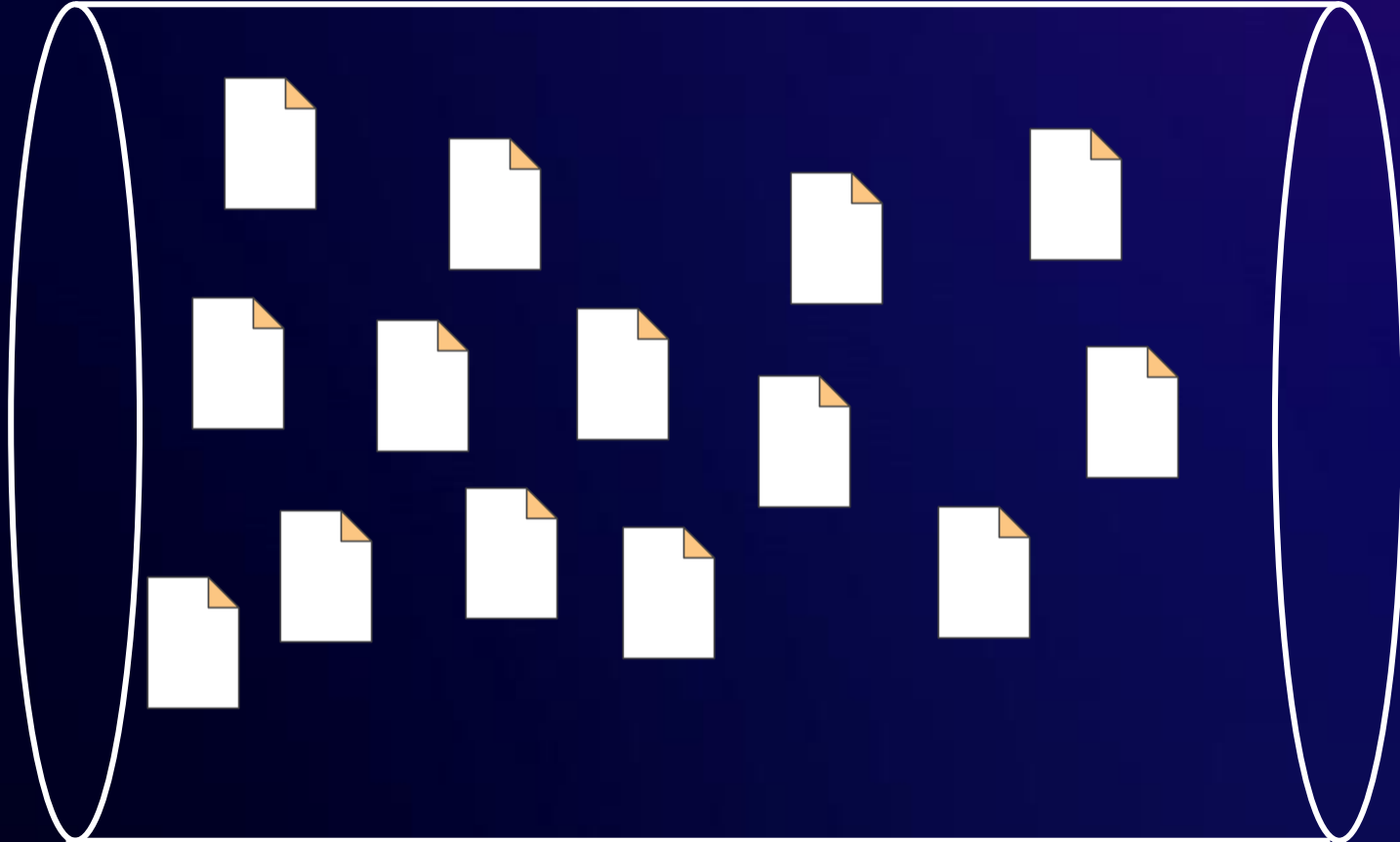
Decoupled applications



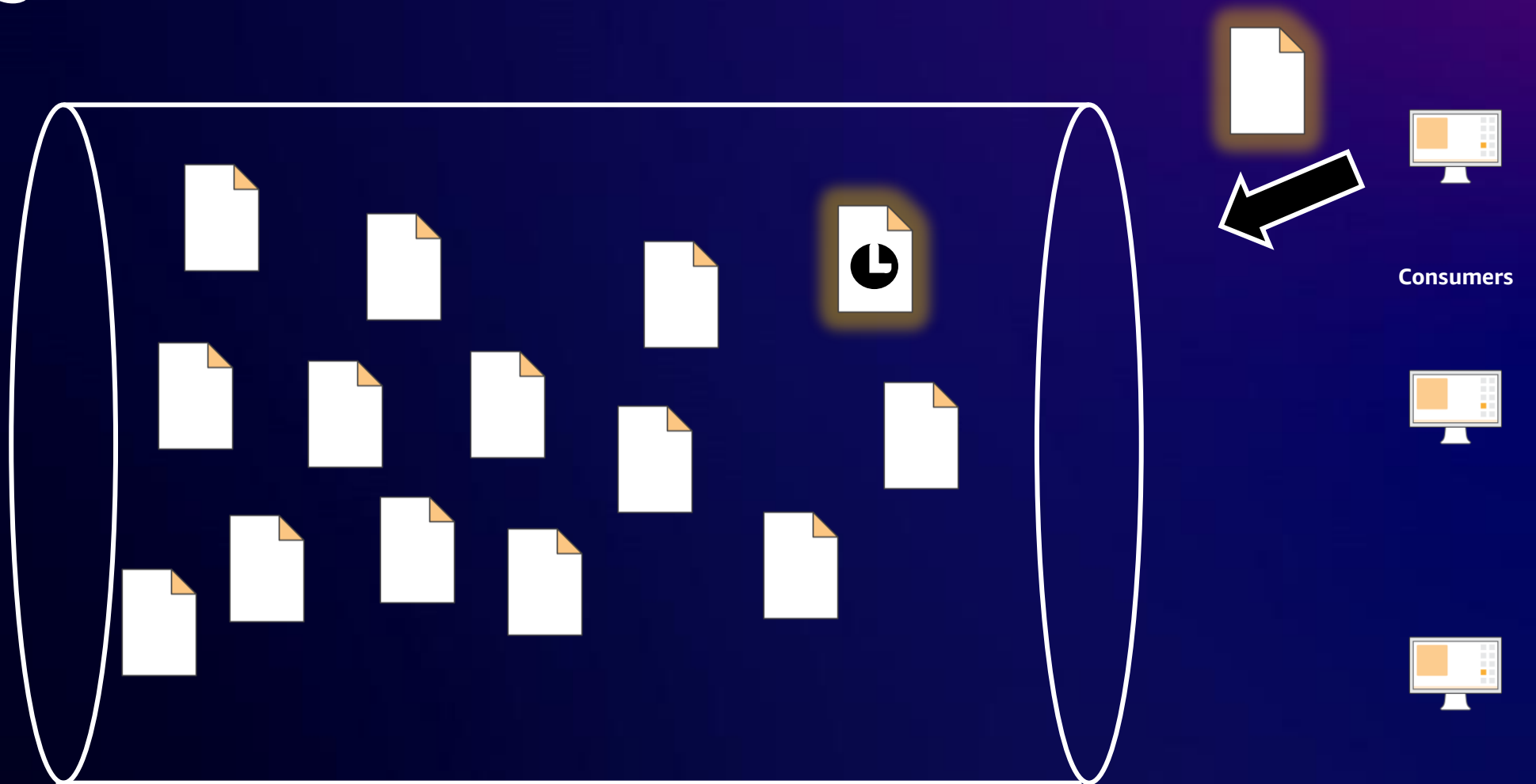
Consuming from Amazon SQS



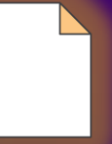
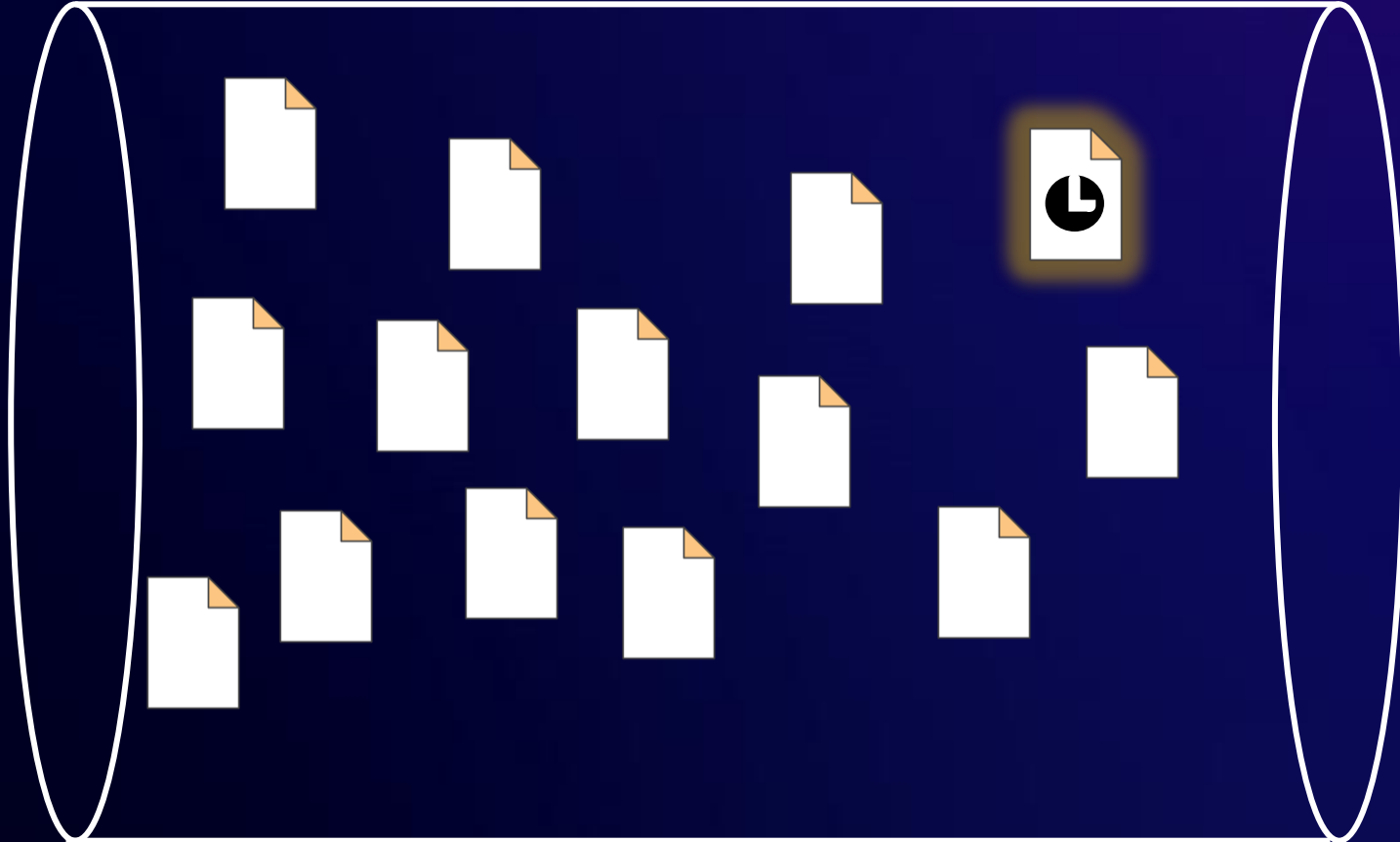
Consuming from Amazon SQS



Consuming from Amazon SQS



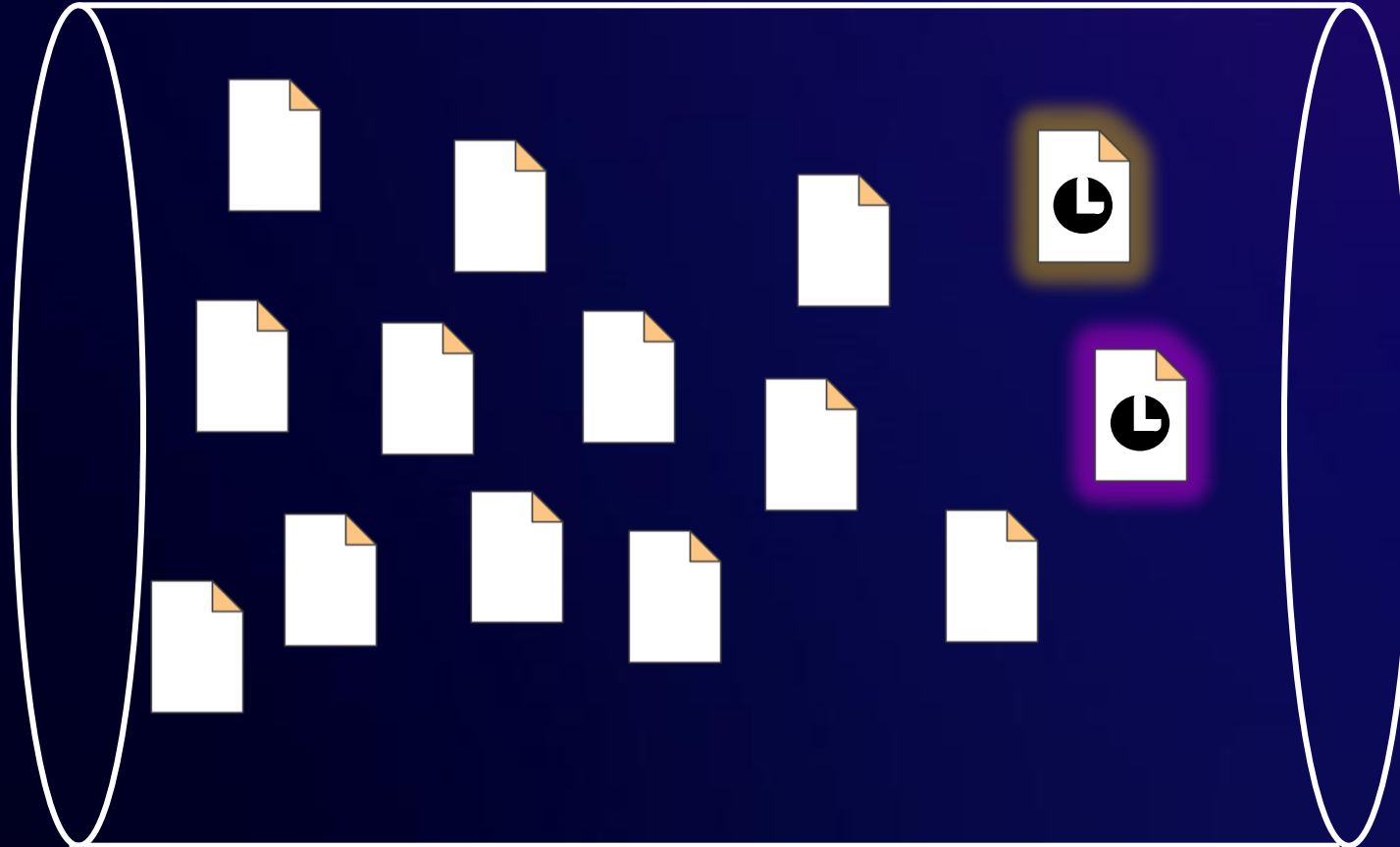
Consuming from Amazon SQS



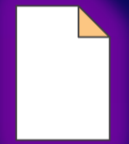
Consumers



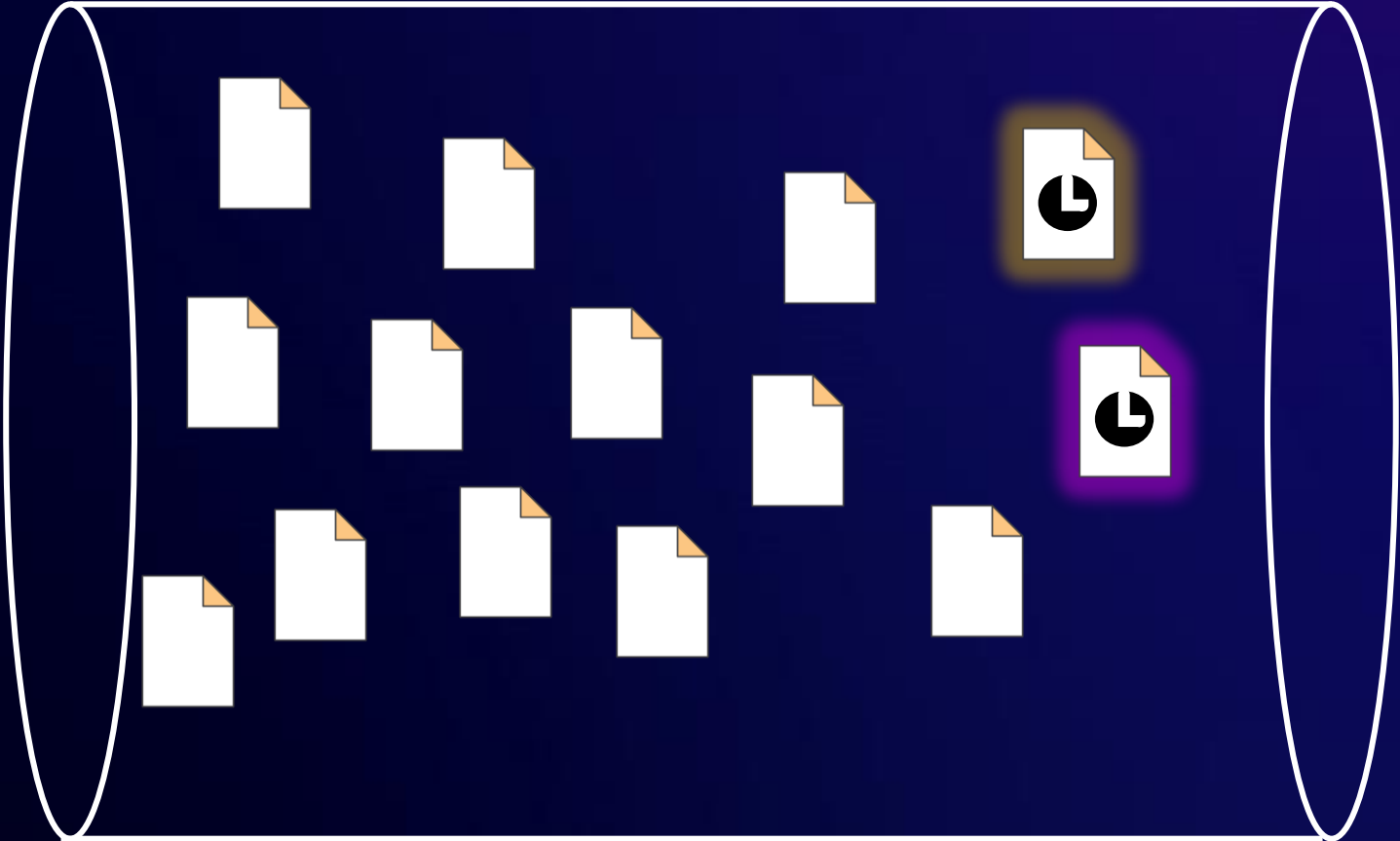
Consuming from Amazon SQS



Consumers



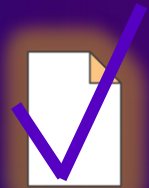
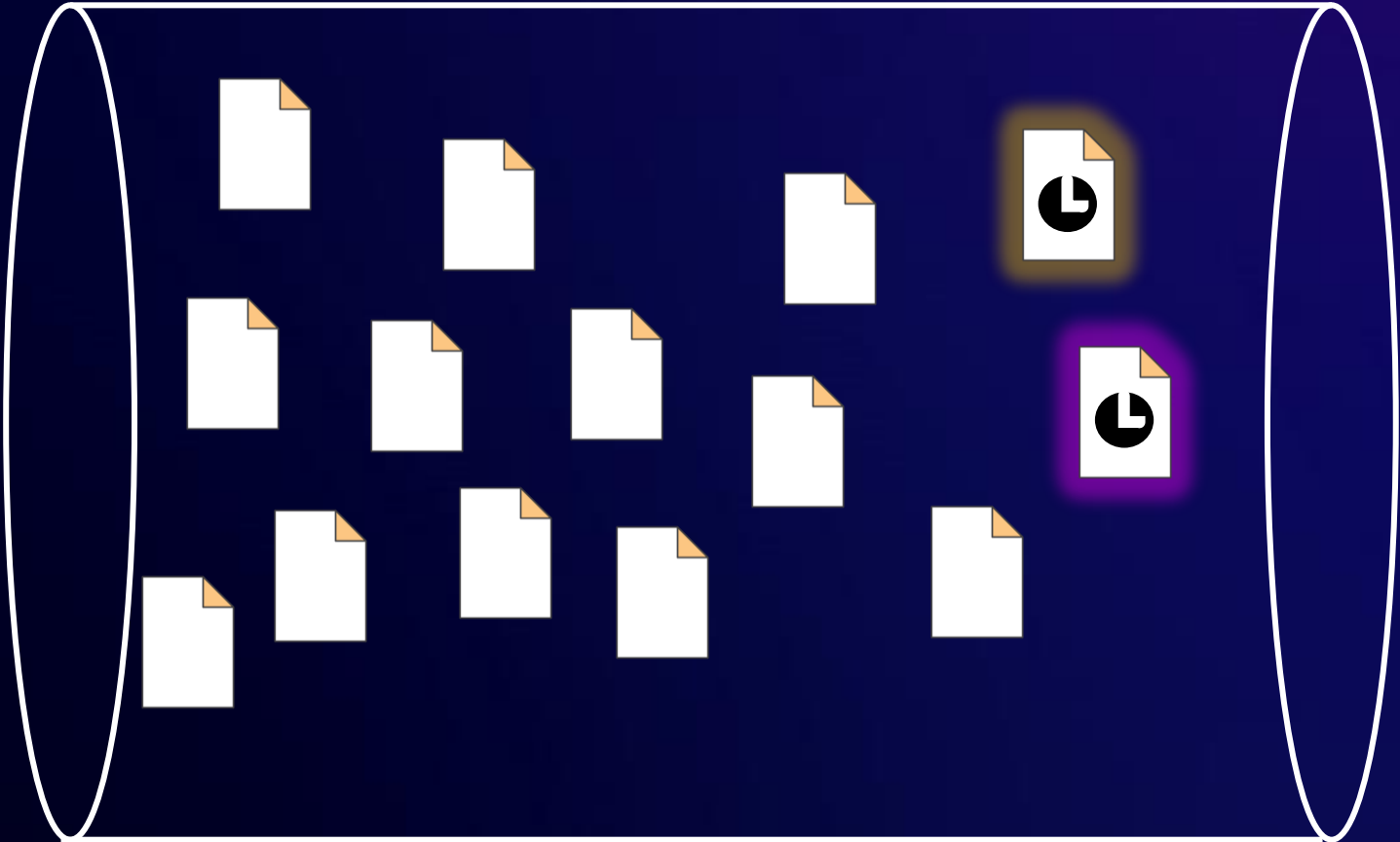
Consuming from Amazon SQS



Consumers



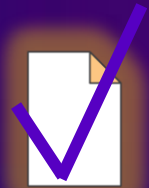
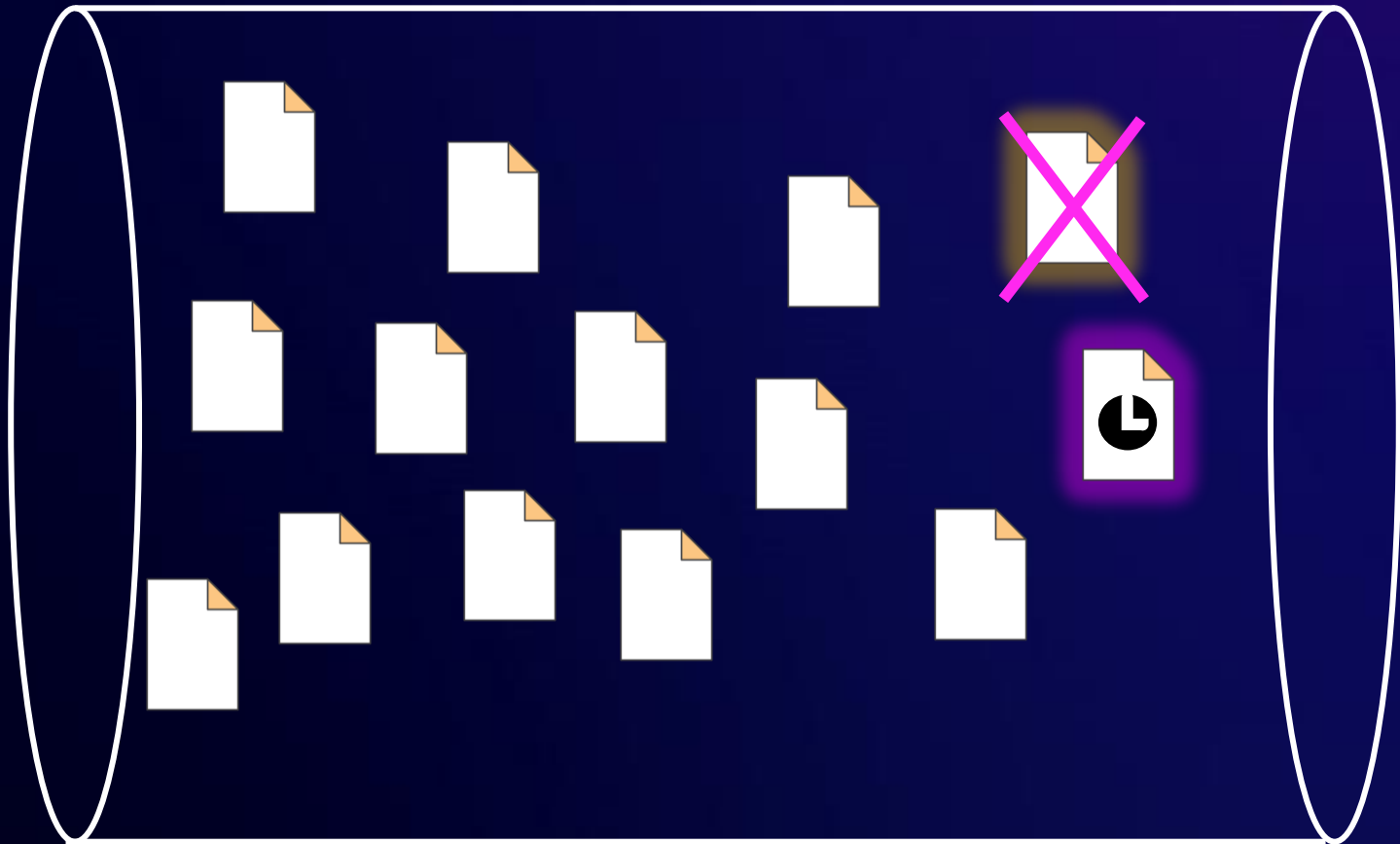
Consuming from Amazon SQS



Consumers



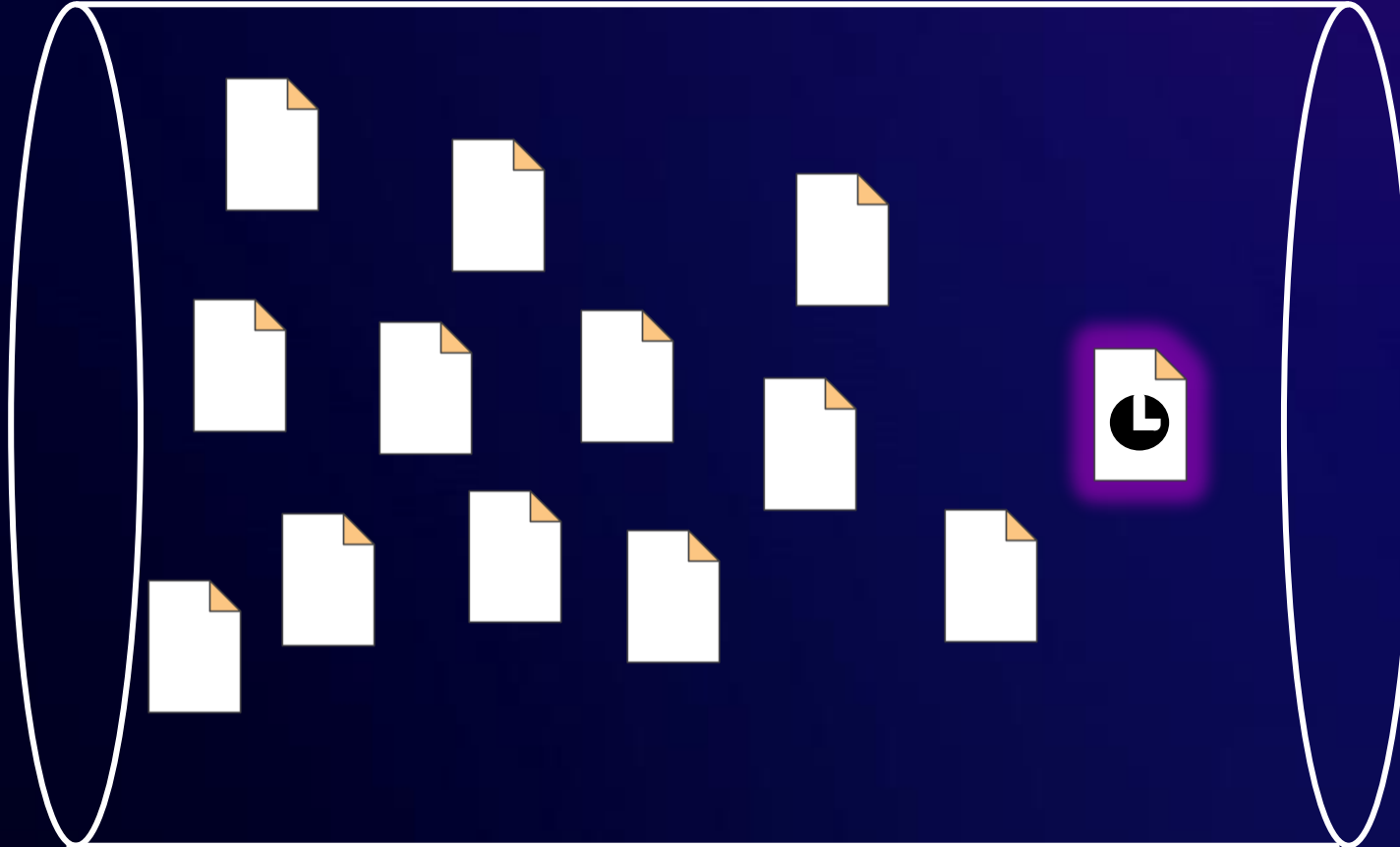
Consuming from Amazon SQS



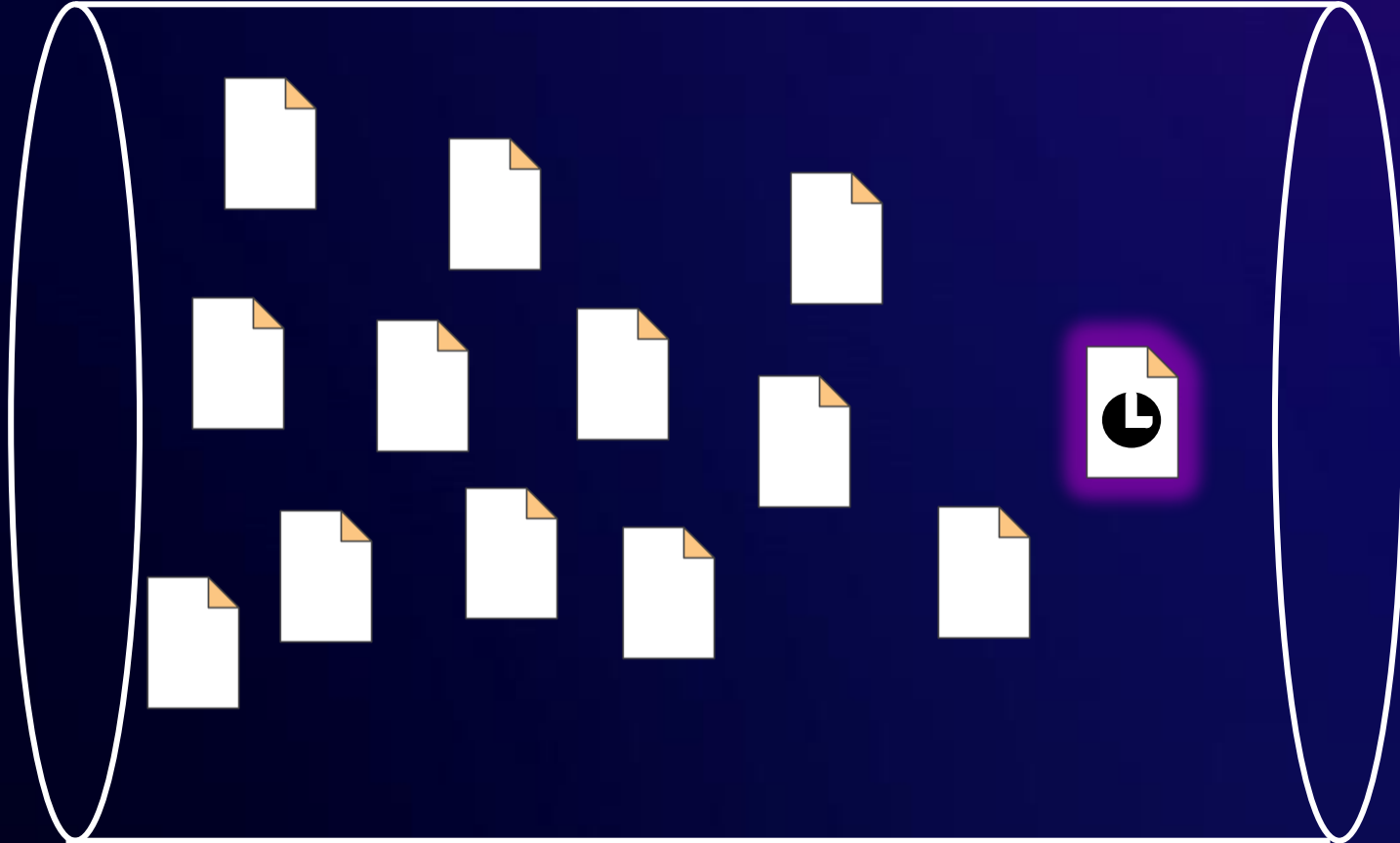
Consumers



Consuming from Amazon SQS



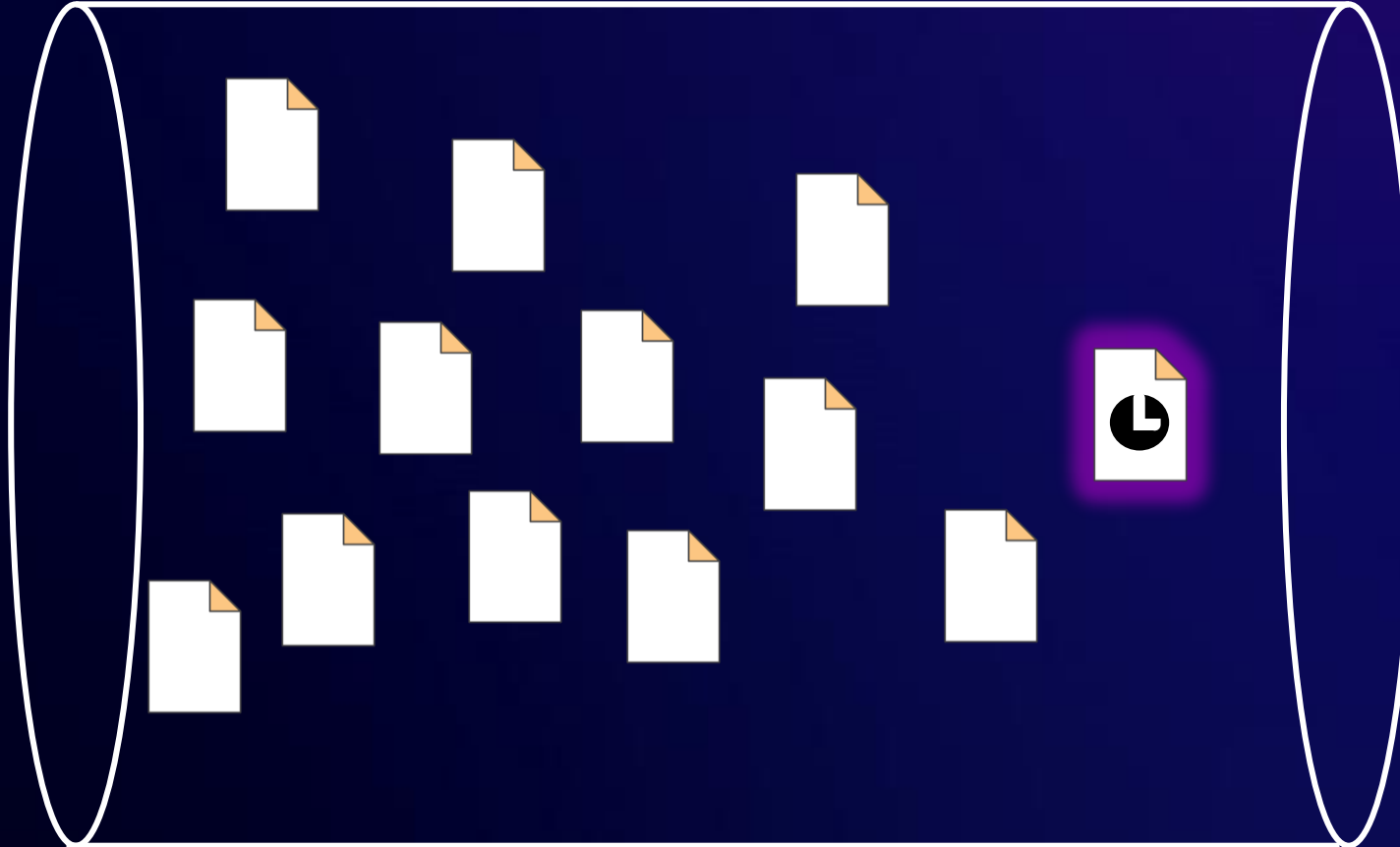
Consuming from Amazon SQS



Consumers



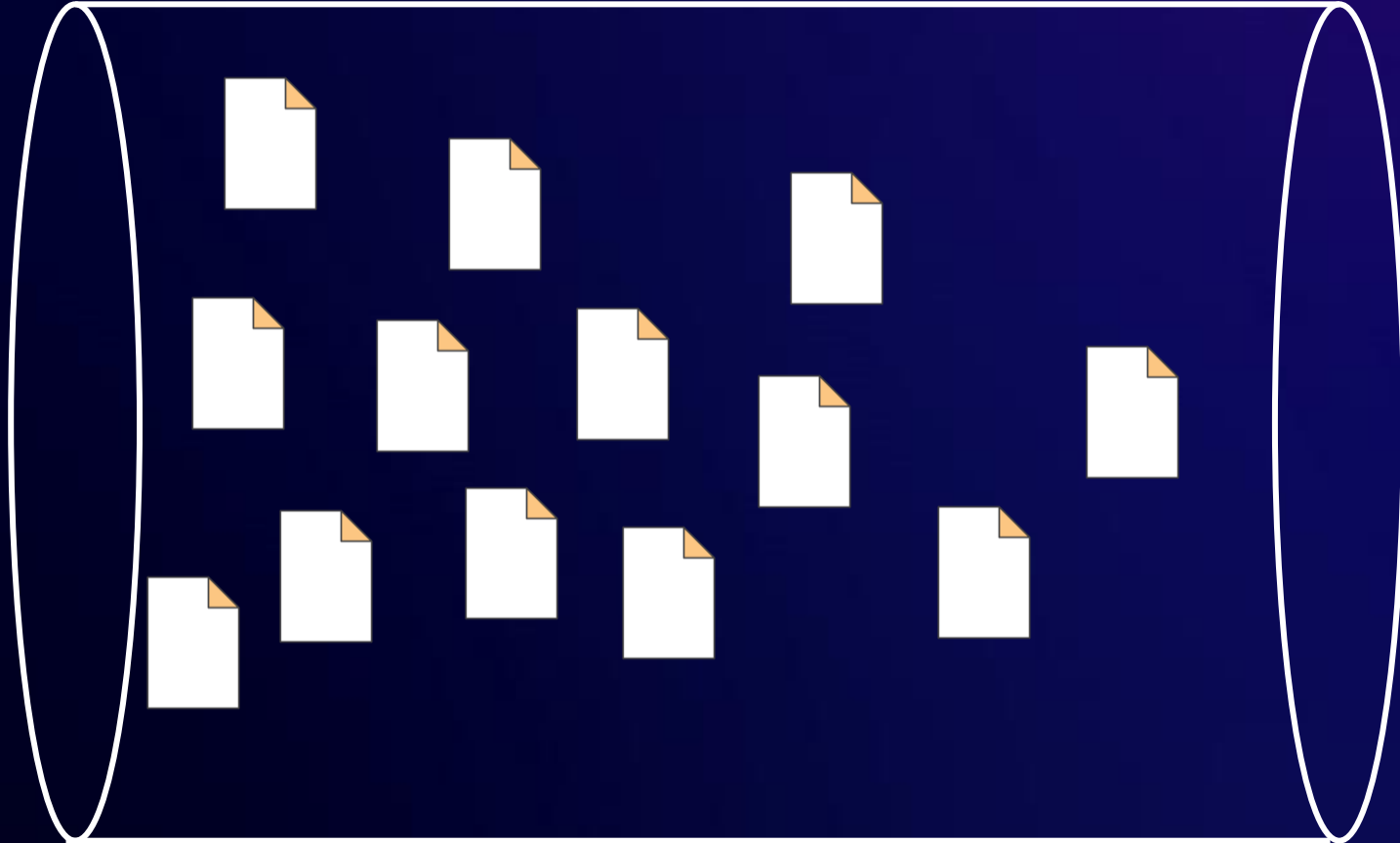
Consuming from Amazon SQS



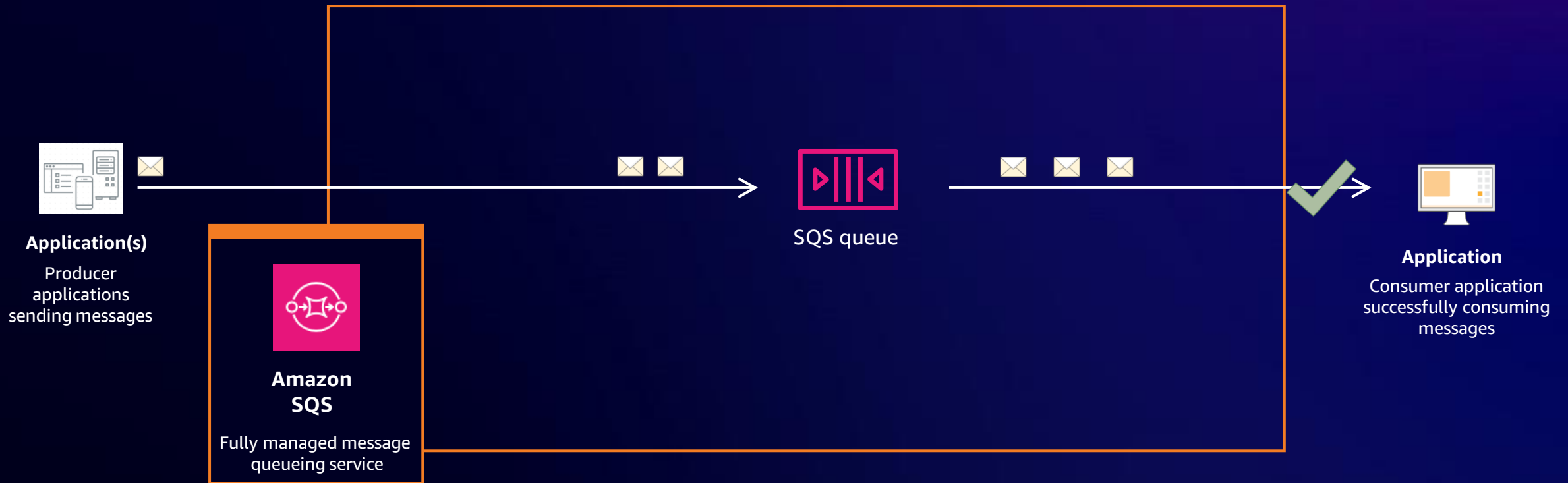
Consumers



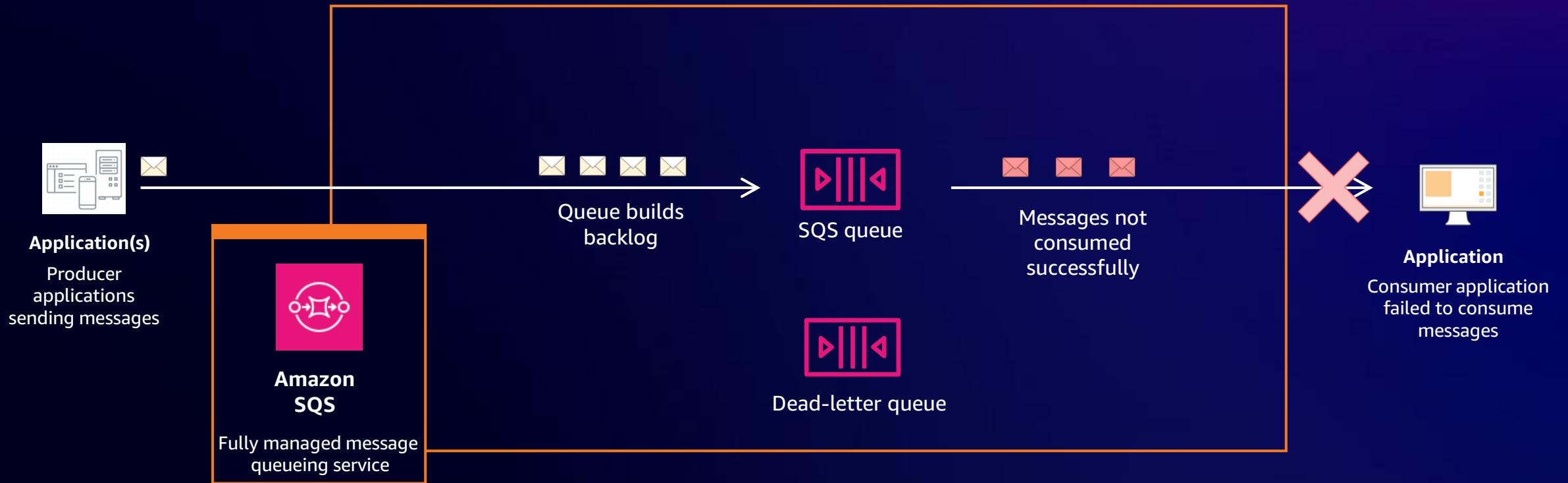
Consuming from Amazon SQS



Lifecycle of successfully consumed messages



Error handling with message processing



Dead-letter queues (DLQs)

DLQs are just like any other Amazon SQS queue where other queues (source queues) can send messages that can't be consumed successfully



**Retry
mechanisms**



**Error
triage**



**Message
safety net**

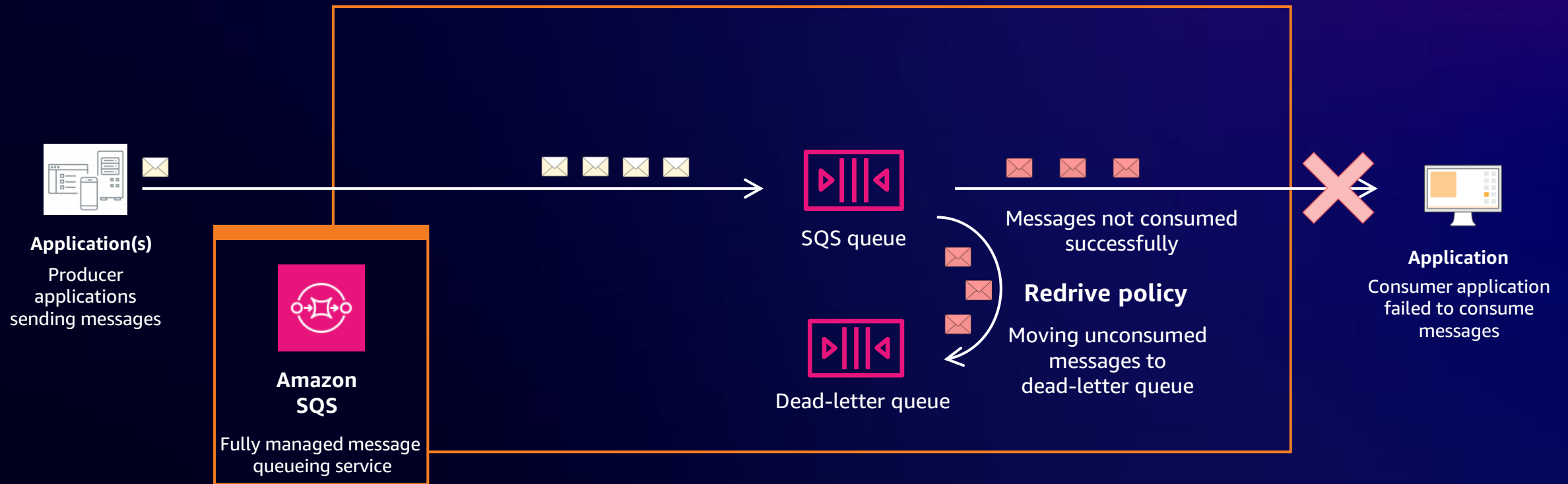


**Supports FIFO
and standard**



**Cost-efficient
troubleshooting**

Error handling using dead-letter queue



Dead-letter queue sample redrive policy

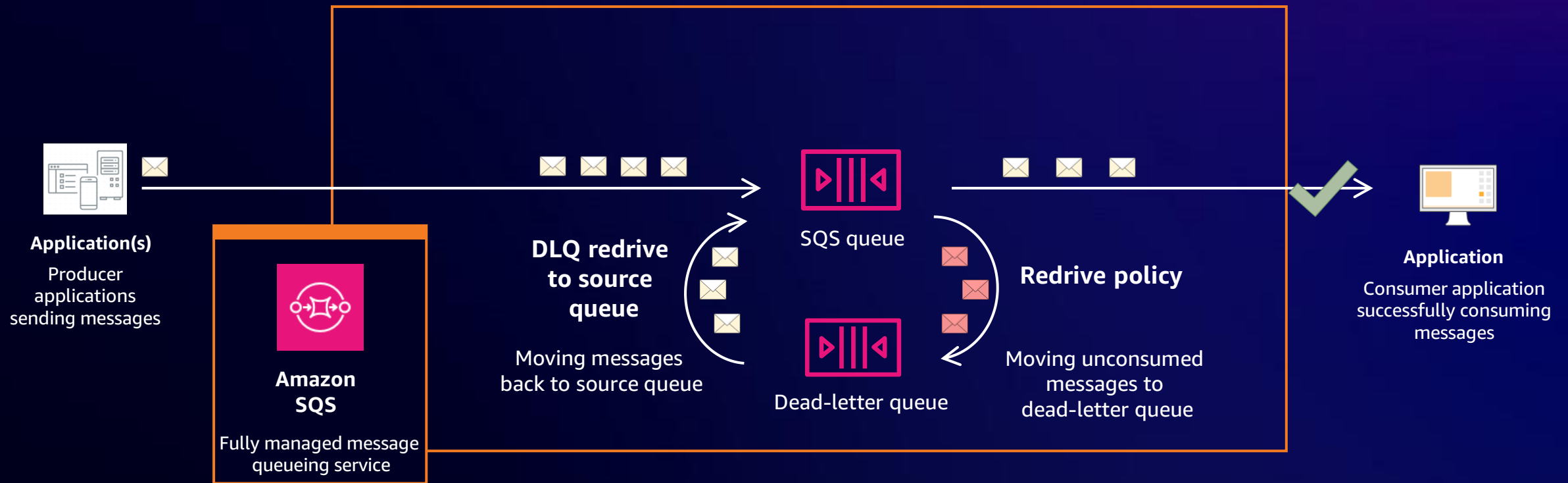
Redrive policy

The *RedrivePolicy* resource accepts the following input properties:

QueueUrl <i>String</i>	The URL of the Amazon SQS queue to which to attach the policy
RedrivePolicy <i>String</i>	The JSON redrive policy for the Amazon SQS queue. Accepts two key/val pairs: <i>deadLetterTargetArn</i> and <i>maxReceiveCount</i>

```
const queue = new aws.sqs.Queue("queue", {});
const ddl = new aws.sqs.Queue("ddl",
{redriveAllowPolicy: queue.arn.apply(arn =>
JSON.stringify({
  redrivePermission: "myQueue",
  sourceQueueArns: [arn],
}))});
const redrivePolicy = new
aws.sqs.RedrivePolicy("redrivePolicy", {
  queueUrl: queue.id,
  redrivePolicy: ddl.arn.apply(arn =>
JSON.stringify({
  deadLetterTargetArn: arn,
  maxReceiveCount: 3,
  })),
});
```

Lifecycle of unconsumed messages



Patterns of dead-letter queues

Crystal Power

The vast, surreal salt flats of Badwater Basin change constantly. Salt crystals expand, pushing the crust of salt into rough, chaotic forms. Newly formed crystals ooze between mudcracks, sketching strange patterns on the surface of the salt flat. Passing rainstorms wash off windblown dust and generate a fresh layer of blinding white salt. Floods create temporary lakes that dissolve salts back into solution, starting the process all over again.

Intense Concentration

The source of Badwater's salts is Death Valley's drainage system of 9,000 square miles—an area larger than New Hampshire. Rain falling on distant peaks creates floods that rush ever lower. Along the way, minerals dissolve from rocks and join the flood. Here, at the lowest elevation, floods come to rest, forming temporary lakes. As the water evaporates, minerals concentrate until only the salts remain. After thousands of years, enough salts have washed in to produce layer upon layer of salt crust.

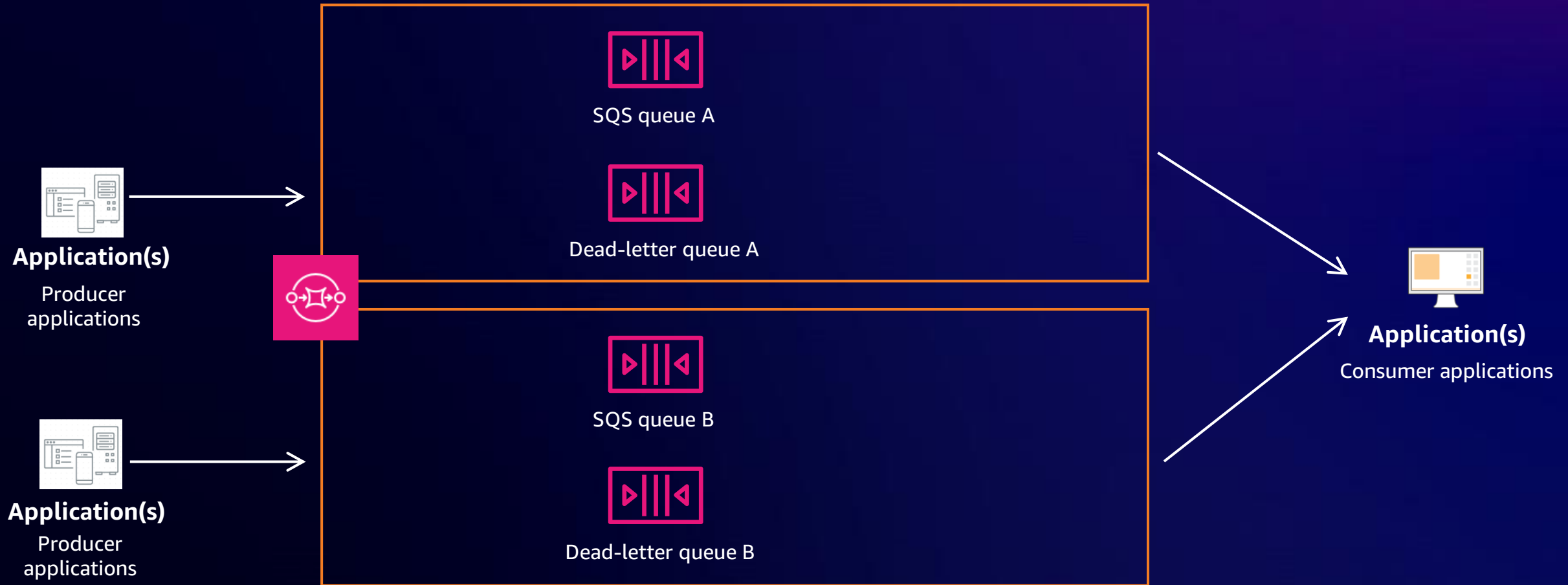
Salt crusts are fragile. Please tread lightly.



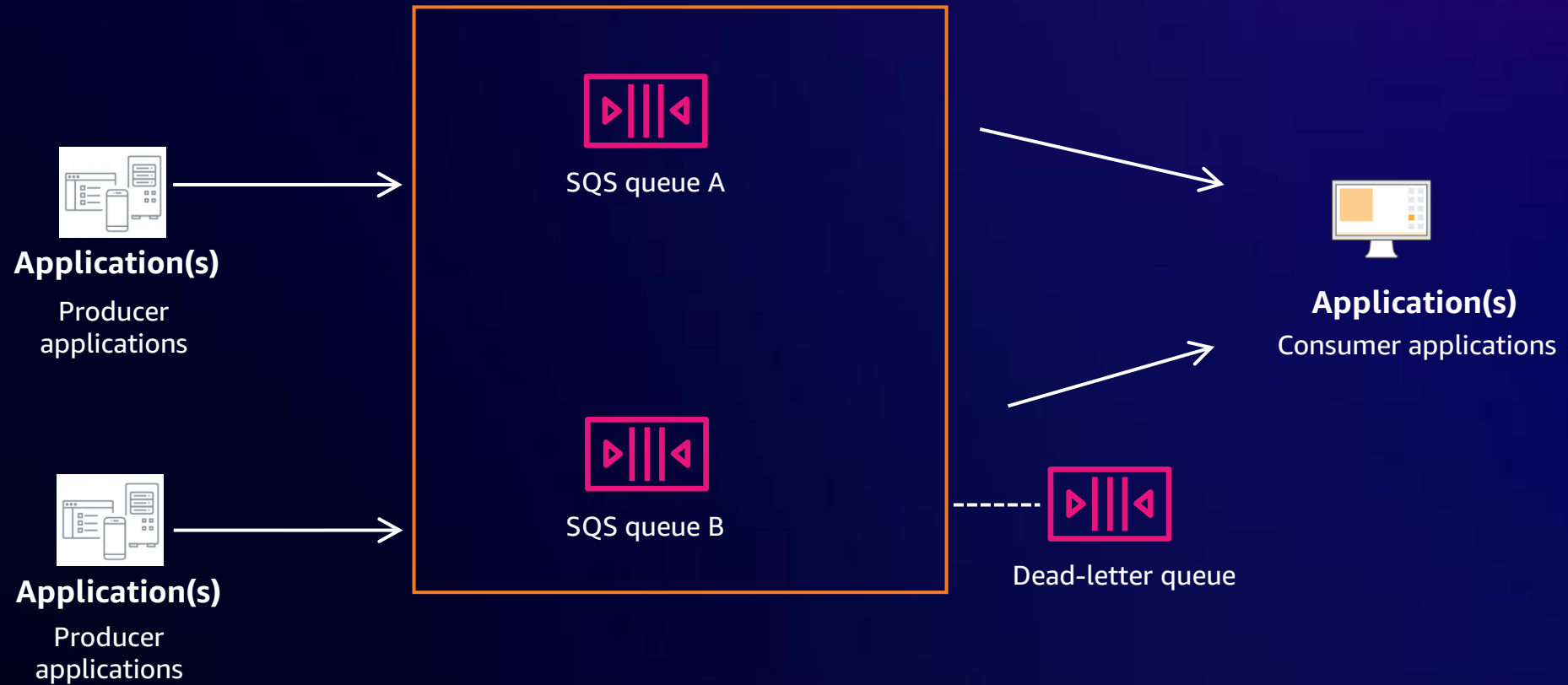
Death Valley National Park
California / Nevada



One-to-one source queue to dead-letter queue



Many-to-one source queues to dead-letter queue



Key takeaways for dead-letter queues

1. **Always enable DLQs for critical queues:** This ensures that failed messages are captured and can be examined promptly.
2. **Regularly review DLQs:** Periodically review the messages in your DLQs to identify patterns of failure.
3. **Implement redrive policies:** Define redrive policies to specify how many times a message should be retried before being sent to the DLQ. This prevents messages from getting stuck in a retry loop.
4. **DLQ retention:** Set the retention period of a DLQ for standard queues to be longer than the retention period of the original queue.
5. **Automated redrive to source queues or custom queues:** Set up automated policies to redrive messages back to source queues or custom queues.

Check out these other sessions

API301 | Building event-driven architectures

Nov. 30 | 11:00 AM – 1:00 PM (PST)

Caesars Forum

API311 | Combine messaging services for workload resilience and ordering

Nov. 30 | 2:00 PM – 3:00 PM (PST)

Caesars Forum

API303 | Navigating the journey to serverless event-driven architecture

Nov. 30 | 2:30 PM – 3:30 PM (PST)

Mandalay Bay



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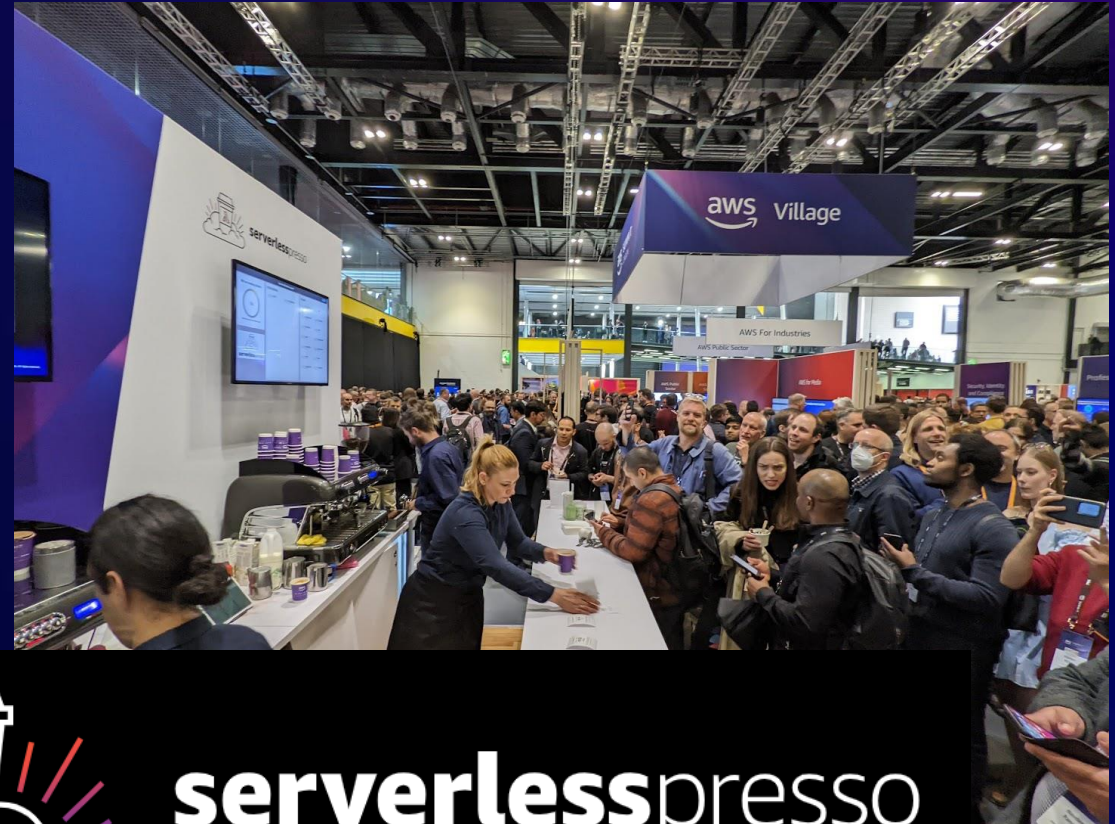


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